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Galvanometer driven optical system for laser dynamic focusing

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Abstract: This paper proposes a novel dynamic focusing module driven by galvanometers to position the laser focus with high speed and high precision. Thanks to the extremely high repeatability and the fast response time of galvanometers, the repeatability of the laser focus positioning is far less than the Rayleigh length of the beam and its response time is around 600 μs. An important feature of the proposed module lies in that it can be further integrated to an XY galvo scanner to realize a 3-axis laser scanning system. Due to the same galvanometers of the dynamic focus module and the XY scanner, the laser focus could be positioned in a simultaneous, rapid and precise manner in all three axes. Various simulation and experimental results demonstrate the feasibility and performance of the proposed dynamic focus module and the processing capability of the 3-axis scanning system with the proposed module.

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1. Introduction

The laser scanning technology is essential in wide applications such as imaging, material processing, 3D printing, etc. Among these applications, it is desirable to position the laser focus along three axes in a high-speed, high-precision and simultaneous manner [1–3], commonly utilizing a 3-axis scanning system composed of an XY scanner and a focus-variable module [4]. The speed and precision of focus positioning are the key factors affecting the throughput and quality in imaging and material processing, and these factors depend on the performance of the XY scanner and the focus-variable module. It is noticed that galvanometers in the XY scanner are able to rotate the reflection mirrors with high precision and high frequency, then the challenge of achieving a high-performance 3-axis scanner mainly lies in the focus-variable system. Conventional methods of focus-control are mainly based on the mechanical movement of optical elements, and the focusing frequency is limited due to the relatively large inertia of lenses and accessories.

Recently, high-speed focus variable optical elements and enhanced adaptive optical technologies have emerged to meet the demand from industrial and scientific communities [3]. With new materials used in tunable focus lenses, liquid crystal-based lenses [5–8] and electro-optic lenses [9–13] can be utilized to change focal length with high speed of several hundred kHz. They, however, tend to be polarization dependent and require high driving voltage. With new principles to manipulate lens shapes, tunable lenses, based on electro-wetting [14–16], dielectro-wetting [17,18] or shape-changeable liquid [19–25], are capable of varying the focus with high speed, while it is difficult to control the curvature of the surface of the liquid lenses in a precision manner [26]. Micro-electromechanical systems (MEMs) such as curvature-variable mirror membranes [27,28] and metasurface lenses [29–31] adjust the focus by means of their deformation, while the limited refocusing response time based on these techniques is relatively slow, usually around 10 ms. Tunable acoustic gradient index (TAG) lens [32–34], driven by acoustic waves generated

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by cylindrical piezoelectric ceramics, can oscillate the focus rapidly with response time less than 1 μs. By converting the linear position of the focus along the optical axis into rotary motion, some optical designs [4,35,36] are utilized to shifting the focus. Specifically, Ref. [4] produces a micro-mirror array for random-access focusing at a response rate of 8.75 kHz with a relatively complicated iterative design process. And Ref. [35] designs a device to achieve dynamic focusing with rapid response time less than 1 ms but should utilize complicated components such as SLM (Spatial Light Modulator) at high cost.

Motivated by the high dynamic performance of galvanometers, here we propose a novel optical design for continuous and fast dynamic focusing module. In specific, two parabolic reflectors and two galvanometers are utilized as shown in Fig. 1(a). The working principle is as follows, with the rotation of the galvanometers, the divergence of the exit beam varies, and hence dynamic focusing is achieved. As the proposed system are comprised of two galvanometers and two parabolic mirrors, the corresponding dynamic focusing module can be rearranged with high flexibility to realize different refocusing ranges. Moreover, as shown in Fig. 1(b), the proposed dynamic focusing module is easy to be integrated with an XY galvanometer to form a 3-axis laser scanning system, and it is feasible to manipulate the laser focus with a high-speed, high-precision and synchronized manner thanks to the four identical galvanometers with the same dynamic performance.

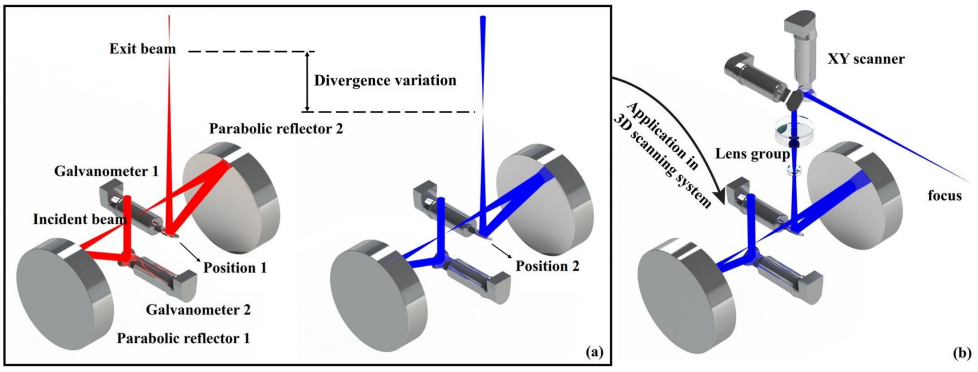

Fig. 1. A conceptual design of a galvanometer driven dynamic focusing system.

The rest of this paper is organized as follows: Section 2 introduces an optical design of the dynamic focusing module and analyzes its features. In Section 3, various experiments are conducted to validate the zooming function of the proposed module and its feasibility in 3-axis scanning, followed by conclusions in Section 4.

2. Optical design

2.1. Novel design of galvanometer driven dynamic focusing module

The proposed optical design of the dynamic focus module is shown in the Fig. 2(a), which is mainly composed of two identical concave parabolic reflectors and plane reflectors. Specifically, two concave parabolic reflectors have parallel axes of symmetry, and the center of the rotating reflectors A and B coincide with the focal point of the concave parabolic reflectors A and B, respectively. The incident beam is collimated, and the incident point on the parabolic mirror A varies according to the rotation of the reflector A. Because different incident points result different focal lengths, the reflector A and the parabolic mirror A can achieve the variation of the beam divergence, thereby realizing the variation of the focal point along the axial direction.

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Note that the optical axis of the laser beam emitted from the parabolic mirror A is not fixed, so it cannot be directly used.

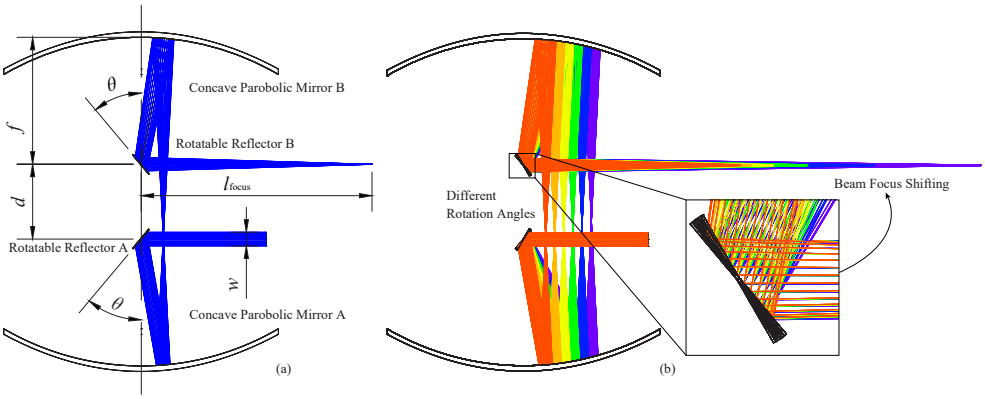

Fig. 2. The principle of the proposed optical design: (a) design parameters of the optical setup; (b) illustration of the beam focus positioning.

To solve the above problem, we adopt the reflector B and the parabolic mirror B according to the feature of parabolic mirrors. It is known that the parabolic mirrors focus the ray parallel to the axis at the focal point, and reflect the ray from the focal point as the ray parallel to the axis. Therefore, after passing through the parabolic mirror B and the plane mirror B, the beam can exit along a fixed optical axis. Then dynamic focusing is realized without changing the outgoing optical axis. Moreover, the parabolic mirrors are coaxially placed to simplify the calculation of the rotation angles.

And to achieve fast motion and synchronize with an XY scanner, the galvanometers with digital servo control are utilized to drive the reflectors at high frequency (>1 kHz) and with high repeatability (<2 μrad), which enables extremely high-speed and high-precision focus positioning.

2.2. *Analysis of variable focusing*

As shown in the Fig. 2(a), parameters d and f are the key to the zooming range and working length of the proposed focus variable module, where f is the parabolic mirror focal length, and d is the distance between two rotatable reflectors.

To demonstrate its dynamic zooming feature, we apply numeric analysis method based on ray tracing strategy since the theoretical optical model of the proposed design is very complicated. Specifically, as shown in the Fig. 2(b), the parameters are set as follows: the parabolic mirror focal length f is 50.8 mm; the distance d between reflectors is 30 mm; the entrance aperture w is 5.5 mm; and the rotating angle θ of the rotatable reflectors is set from 30° to 40°. It is observed from Fig. 2(b) that different rotating angles θ correspond to different positions of the laser beam focus.

Furthermore, to analyze how the two parameters affect the dynamic focusing features of the proposed optical design, the focus shifting range and working focal length with different f and d are calculated. The set parameters of calculation are as follows, the input rotation angle of the rotatable reflectors θ is set as 30° and 40°, and the incident beam aperture w is 5.5 mm, and the focal length of parabolic mirrors f and the distance d between reflectors are set from 30 mm to 80 mm, respectively. Two methods are utilized to analyze the focus position of the proposed optical system, by which are ray tracing based on geometrical optics and wavefront propagation based on physical optics. The simulation is conducted based on the commercial

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software package ZEMAX. The simulation model is shown in Fig. 3, and the simulation results of energy distribution under different design parameters are obtained by the physical optics propagation of the Gaussian beam.

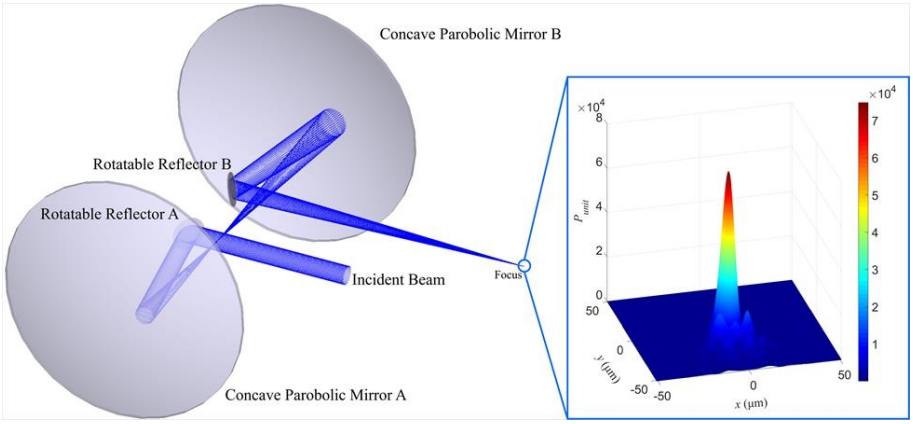

Fig. 3. Simulation model of the proposed optical design.

The relationship between the focal length l_{focus} and the two parameters is shown in Fig. 4(a) and Fig. 4(b), it is observed that the l_{focus} increases as f increases or d decreases. And Fig. 4(c) shows the relationship between the focus zooming range Δl_{focus} and the two parameters, the Δl_{focus} increases as f increases or d decreases.

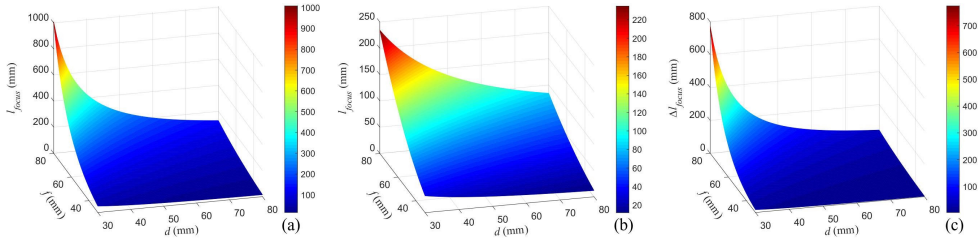

Fig. 4. Results of the theoretical analysis: (a) variations of l_{focus} with f and d ($\theta = 30°$); (b) variations of l_{focus} with f and d ($\theta = 40°$); (c) variations of Δl_{focus} with f and d (θ varies from 30° to 40°)

With the above numeric calculation results, an interpolation method is applied to select the appropriate parameters and the varying range of the rotating angle θ towards different refocusing requirements. It means a dynamic focusing module with different working length and focus zooming range could be conveniently designed, by choosing the parabolic mirrors with different focal length f and by adjusting the distance d between the two galvanometers.

2.3. Application to 3-axis scanning

To improve the beam focus shape and to adapt the aperture of output laser beam, a lens group is integrated in the proposed optical design as shown in Fig. 5. The lens group functions as beam expansion and focusing, and the simulation results show an improved beam focus shape compared with the focus shape in Fig. 3 (right).

For the application of 3-axis laser scanning, the proposed dynamic focusing module is adopted with an XY galvanometer scanner as shown in the Fig. 6(a). And the further simulation results

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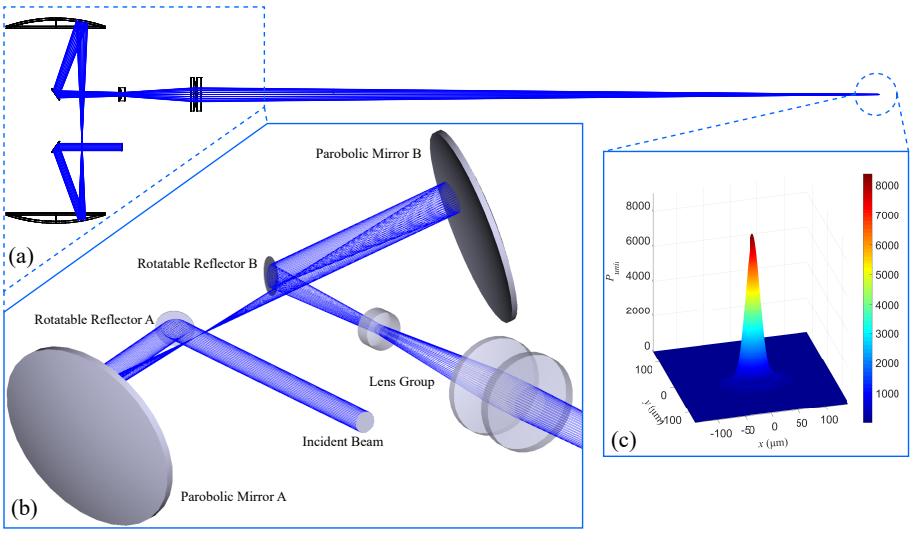

Fig. 5. Simulation of dynamic focusing module with a lens group: (a) the overall simulation model; (b) parts in the simulation model; (c) simulation results of irradiance distribution at the focus position (physics optical propagation of Gaussian beam).

are shown in the Fig. 6(b), which indicate that the laser focus is positioned in three dimensions by varying the rotating angles of four plane reflectors.

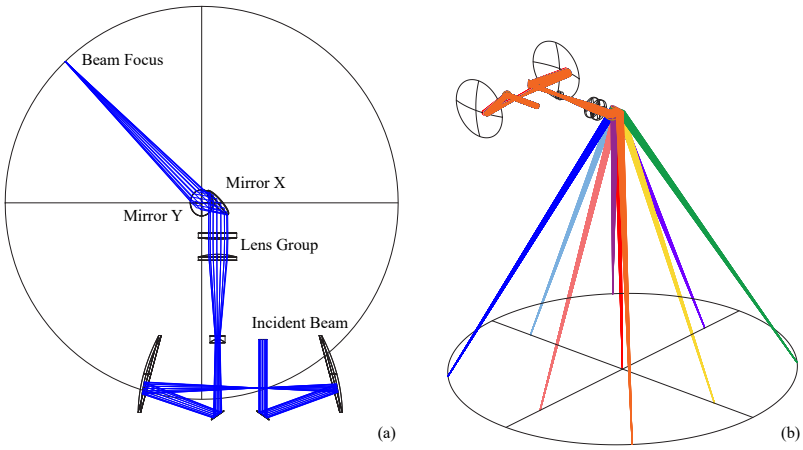

Fig. 6. Schematic of a 3-axis laser scanning system based on the proposed dynamic focusing system: (a) optical design for 3-axis laser scanning; (b) the simulation result of 3-axis scanning.

The relationship between the position of laser focus and rotating angles is shown in Fig. 7. It is noted that the distance between the center points of two mirrors could be decreased by tilting the rotation axis of mirror X, and the required size of mirror Y could be reduced.

The mirror X rotating angle θ_x and mirror Y rotating angle θ_y can be derived as

$$\theta_x = \frac{1}{2}\arctan\left(\frac{x}{e+\sqrt{d^2+y^2}}\right), \tag{1}$$

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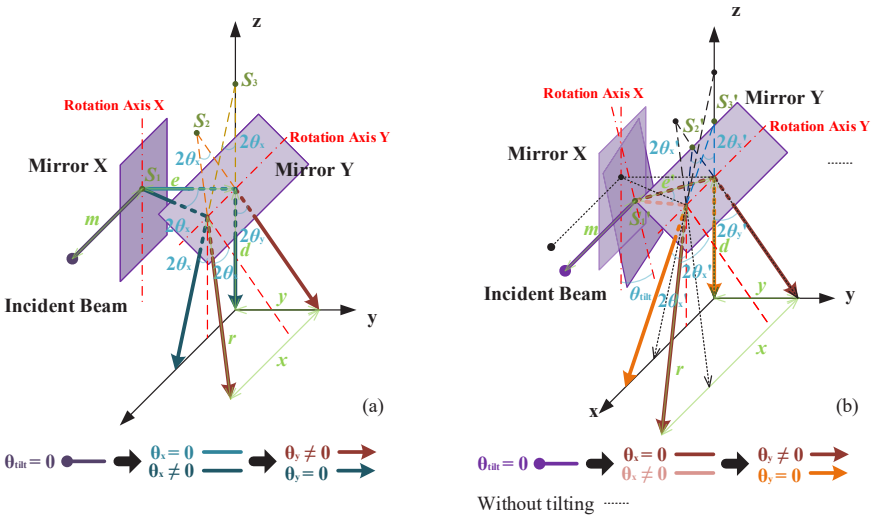

Fig. 7. Schematic of the relationship between the scanning point and mirror angles: (a) before tilting the rotation axis of mirror X; (b) after tilting the rotation axis of mirror X.

$$\theta_{\mathrm{y}} = \frac{1}{2}\arctan\left(\frac{y}{d}\right), \tag{2}$$

where (x, y, d) are the coordinates of laser focus position, and e is the distance between the center points of two mirrors, which is affected by the tilting angle θ_{tilt} of mirror X.

Moreover, the proposed dynamic focusing module is able to change the working length $l(\theta_{\mathrm{z}})$. And the desired working length $l(\theta_{\mathrm{z}})$ can be obtained as

$$\begin{aligned} l(\theta_{\mathrm{z}}) &= e + m + r \\ &= e + m + \frac{x\sqrt{d^2 + y^2}}{e + \sqrt{d^2 + y^2}}, \end{aligned} \tag{3}$$

where θ_{z} is the rotating angle of mirrors within the proposed dynamic focus module, and r is the working length of the scanning system, and m is the distance between the incident aperture and mirror X. Then θ_{z} can be determined from the focus position coordinates of the desired scanning trajectory through (3).

3. Experimental study

3.1. Experiment setup

As shown in Fig. 8(c), the proposed dynamic focusing module is prototyped. The collimated incident beam is 405 nm continuous laser with the diameter of 5.5 mm, passing through two parabolic mirror (ϕ50.8×54.45 mm EFL 30° OAP Protected Aluminum Mirror) and two rotation mirrors driven by self-developed galvanometers with the incident aperture of 20 mm. Reflected by above mirrors, the beam passes through the lens group consisting of lens A ($f_{\mathrm{A}} = -15$ mm), lens B ($f_{\mathrm{B}} = 100$ mm) and lens C ($f_{\mathrm{C}} = 200$ mm).

The experiment setup shown in Fig. 8(a) and Fig. 8(b) is utilized to demonstrate the dynamic focusing performance. The setup consists of the prototype of the proposed dynamic focusing module, a laser source (405 nm, diode laser, continuous wave), a laser interferometer (Attocube IDS3010), a beam profiling camera (Spiricon SP620, Ophir), a vertical lift stage and a host computer. The beam profiling camera, the vertical lift stage and the laser interferometer

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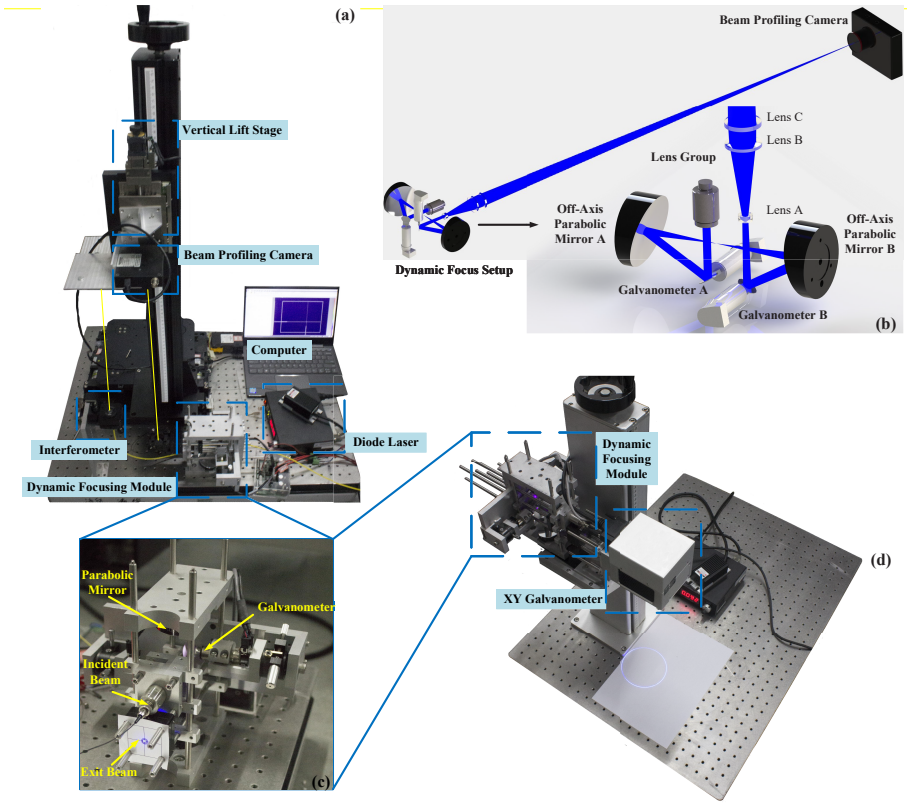

Fig. 8. (a) the first experiment setup demonstrating the dynamic focusing performance; (b) schematic of the first experimental setup; (c) prototype of dynamic focusing module; (d) the second experiment setup for 3-axis laser scanning.

are adopted to measure the irradiance distribution and position variation of the laser focus. Furthermore, the dynamic focusing module is assembled with a self-developed XY scanning head to achieve a 3-axis scanning system as shown in Fig. 8(d).

3.2. *Experiments on dynamic focusing features*

With the above setup, we conduct numerous experiments to demonstrate the focus zooming features and processing capability of the proposed optical system. With the given angles of the galvanometers in the dynamic focusing module, the beam profiler measures the relative irradiance distribution of the beam transverse profile at certain frame rate. Meanwhile, the camera is moved along the vertical direction of the focused beam with the stage. And the laser interferometer is utilized to measure the relative displacement of the beam profile camera.

The measurement results are shown in Fig. 9(m), in which θ_b is the rotating angle of the laser beam and Δl_{focus} is the relative change of focus position. It is observed that Δl_{focus} has good linearity with θ_b, which is consistent with the analytical result. Figure 9 also shows the irradiance distribution of the laser beam at the focus, indicating that the focused beam spot keeps a relative constant shape in different positions along optical axis and the geometric aberrations in the system have negligible effect on beam quality.

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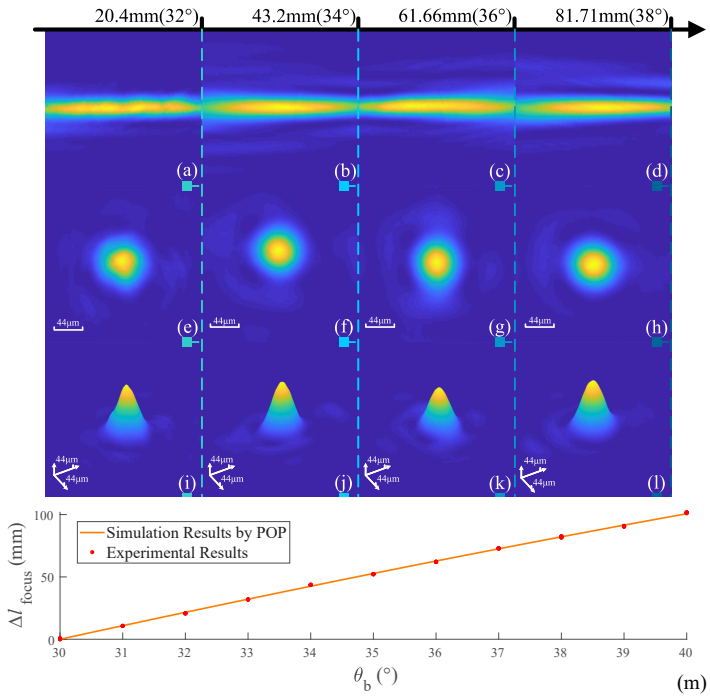

Fig. 9. Experimental results of the focus measurement: (a, b, c and d) the axial irradiance distributions at the beam focus with different rotation angles; (e, f, g and h) the lateral irradiance distributions in the respective focal plane with different rotation angles; (i, j, k and l) the corresponding isometric view of the irradiance distributions with different rotation angles; (m) the comparison between the simulation results and experimental ones of the focus position.

3.3. *Experiments on laser scanning with dynamic focusing module*

Laser scanning experiments are carried out with the scanning system in Fig. 8(d) and the details are as follows. With the given angles of galvanometers in the dynamic focusing module, the system scans a circular trajectory on the plane as shown in Fig. 10 (left). After lifting the system by a certain height, an unclear circular trajectory is scanned on the same plane due to the laser defocusing. Then, with focus control of the laser beam by the dynamic focusing module, the scanning system realized a circular trajectory which is consistent with the trajectory before lifting.

Moreover, a series of lithography processing experiments are conducted with resin coated substrates. The manufactured result is shown in Fig. 10 (right), and it is observed that the processing result keeps a relative constant line width.

3.4. *Experiments on the dynamic performance of galvanometers*

To demonstrate the dynamic performance of the galvanometers in the proposed dynamic focusing module, we conduct various of experiments to measure the step response time and repeatability (root-mean-square value). The references consist of a step signal with amplitude of 3.5 mrad (1% of full scale) and a series of ramp signals from 32 mrad to different positions and back to 32 mrad. The experimental results are shown in Fig. 11, the step response time is 600 μs and the repeatability is 0.4 μrad.

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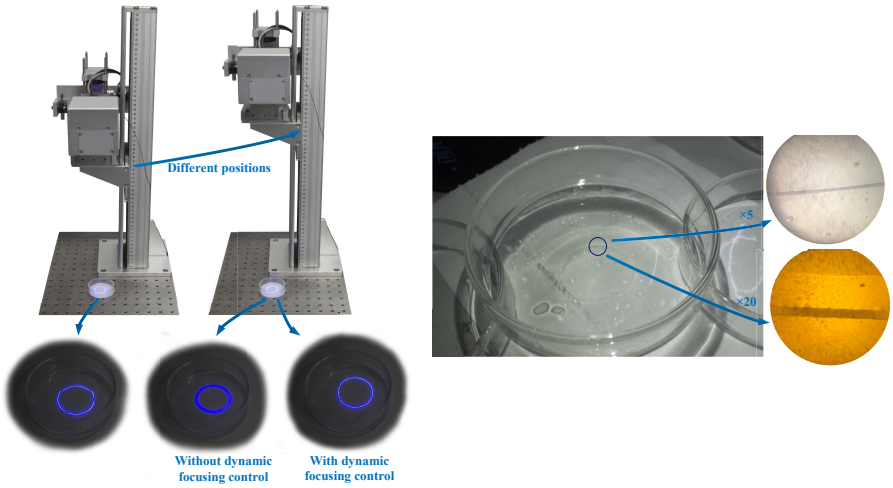

Fig. 10. (left) experimental demonstration of the zooming feature; (right) the lithography processing result.

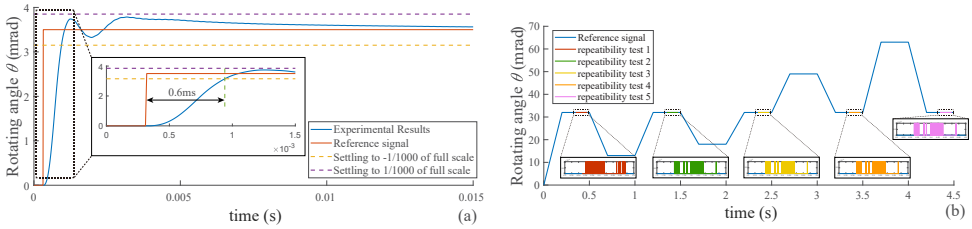

Fig. 11. Experimental results of the dynamic performance of the galvanometers: (a) results of the step response experiment; (b) results of the repeatability experiment.

Due to the linearity between the focus shifting range and beam rotating angle as shown in Fig. 9, the repeatability ε_f of focus positioning can be obtained by

$$\varepsilon_f = \frac{360^\circ \cdot \Delta l_{focus}}{\pi \cdot \Delta\theta_b} \varepsilon_g, \tag{4}$$

where $\Delta\theta_b$ is the variation of the rotating angle of beam (8°), and Δl_{focus} is the relative focus shifting range (81.71 mm) and ε_g is the repeatability of the galvanometers (0.4 μrad). So the repeatability of focusing positioning (468 nm) is far less than the range of the Rayleigh length, and focus positioning in the range of 2.04 mm (equal to the rotation range of 3.5 mrad) can be reached within 600 μs.

4. Conclusions

A novel design of laser dynamic focusing module driven by galvanometers is proposed and prototyped, and various experiments are conducted to validate its feasibility and dynamic performance. The simulation and experimental results show that the proposed system is able to achieve high-speed and high-precision laser focus-positioning with a response time of 600 μs and a repeatability (468 nm) far less than the Rayleigh Length of focusing beam. Moreover, thanks to the same galvanometers in the dynamic focusing module and the XY scanner, the proposed system could achieve 3-axis laser scanning with a high-speed, high-precision and simultaneous manner in all three dimensions. In comparison with conventional focus-control methods, the

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proposed dynamic focusing module offers an effective and convenient approach to achieve a high-performance 3-axis laser scanning system.

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Data availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

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